

RAJPAL SINGH RATHORE

Software Development Engineer

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PROFESSIONAL SUMMARY

Software Development Engineer with hands-on experience in building and deploying end-to-end systems across the full SDLC — from architecture design and backend development to REST API integration, automated testing, and CI/CD deployment. Proven track record of driving quality, reducing manual effort, and delivering production-grade solutions in Agile environments. Strong foundation in OOP, system design, version control (Git), and cloud deployment (AWS EC2). Eager to contribute technical expertise and mentor peers in a collaborative, quality-focused engineering team.

EXPERIENCE

AI Automations Engineer Intern → SDE | TaskNovaMay 2025 – Present · Pune, India

End-to-end software development across the full SDLC — design, development, testing, deployment, and iteration:

- Owned the full SDLC for 10+ production systems — from requirements gathering and architecture design to code review, testing, deployment on AWS EC2, and post-launch monitoring using structured logging and error-tracking strategies.
 - Designed and developed 150+ backend automation workflows integrating REST APIs, webhooks, and function-calling — applying OOP principles and modular code architecture to ensure maintainability and scalability.
 - Built comprehensive test strategies for AI pipelines — including unit tests, integration tests, and end-to-end validation — reducing post-deployment defects and manual QA effort by 80%.
 - Integrated 20+ third-party services (OpenAI, Supabase/PostgreSQL, Slack, Apify, Airtable) via REST APIs and JSON/XML data contracts, implementing fallback logic and schema validation for system reliability.
 - Collaborated in an Agile team environment — participating in sprint planning, code reviews, and retrospectives — while contributing to technical documentation including architecture diagrams, API specs, and integration guides.
 - Supported CI/CD deployment pipelines on AWS EC2 using Git-based version control and Docker containerization, enabling reliable, repeatable deployments with rollback capability.
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PROJECTS

Revenue Intelligence Platform — Multi-Source Data Pipeline · TaskNovaMar 2026

Production-grade backend system for ingesting, normalizing, and querying business data from multiple sources.

- Designed and implemented a canonical data pipeline in PostgreSQL (Supabase) ingesting data from Gmail, Exotel (calls), Fathom (meetings), and CRM — with per-source normalization, identity resolution, and deduplication logic.
- Applied robust system design principles — webhook-driven event ingestion, schema validation via Code nodes, and Switch-based routing — ensuring correctness, reliability, and debuggability across all data paths.
- Authored complete technical documentation: architecture diagrams, data schema definitions, webhook payload contracts, and integration runbooks used by the team.

AI Calls Q&A Intelligence — Large-Scale Call Processing Platform · TaskNovaDec 2025

- Built an end-to-end call intelligence pipeline in Python processing 1,000+ sales/support calls — covering transcription, sentiment analysis, and objection detection — with structured test coverage at each pipeline stage.
- Implemented a document-to-vector embedding pipeline enabling semantic search across company documentation, improving audit accuracy and reducing manual quality analysis effort by 80%.
- Designed and maintained automated evaluation scripts to benchmark pipeline output quality and flag regressions, enabling continuous improvement of AI-generated insights.

SuperBot — Natural Language Database Interface · TaskNovaSep 2025

- Built an LLM-powered system converting natural language to SQL — applying OOP design patterns, schema embedding for context-aware query generation, and rigorous input validation — improving query accuracy by 90%.

- Implemented automated test suites validating SQL output correctness across diverse query types; tracked defect patterns and iterated on prompt engineering to resolve root causes systematically.

Hey Buddy — AI Vision & Voice Assistant (Award-Winning) · College Exhibition2024

- Developed a modular computer vision and voice AI system — selected as 1st place among 50+ teams — demonstrating strong fundamentals in OOP, system integration, and delivering high-quality software under constraints.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS

Backend & APIs: FastAPI, REST APIs, JSON, XML, Webhooks, Function Calling, Fallback Logic

OOP & Design: Object-Oriented Design, System Architecture, Modular Code Design, Design Patterns

Testing: Unit Testing, Integration Testing, System Testing, Automated Test Scripts, TDD principles

DevOps & Cloud: Git, Docker, AWS EC2, CI/CD Pipelines, Deployment Automation, Performance Monitoring

Databases: PostgreSQL (Supabase), Vector Databases, pgvector, SQL Query Optimization

AI/ML Systems: LLM Pipelines, RAG Architectures, Embeddings, Agentic AI, Multi-Agent Systems, Prompt Engineering

Tools & Platforms: n8n, Make, Airtable, Notion, Slack, Apify, OpenAI API, Docker

Methodologies: Agile/Scrum, SDLC, Code Review, Technical Documentation, Architecture Diagrams

EDUCATION

Bachelor of Business Administration (Computer Applications) | Savitribai Phule Pune University2022 – 2025 · Pune, India

Core Coursework: Software Engineering, Data Structures & Algorithms, DBMS, Object-Oriented Programming, System Design

Full Stack Java CertificationCapgemini & Symbiosis Skills University

ACHIEVEMENTS & RECOGNITION

- 1st Place — College Project Exhibition: Awarded among 50+ competing teams for end-to-end AI system design, demonstrating technical excellence and effective communication of complex software to a non-technical panel.
- Built and maintained 10+ production-grade AI systems at TaskNova, each going through complete SDLC — design, review, test, deploy, and monitor — within an Agile sprint structure.